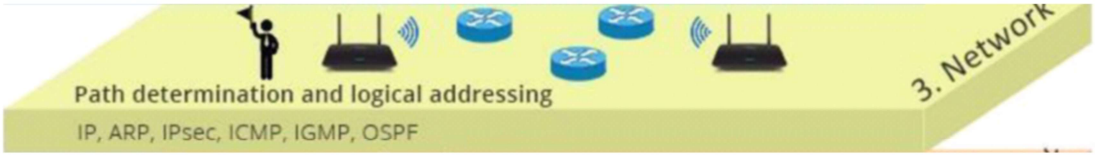
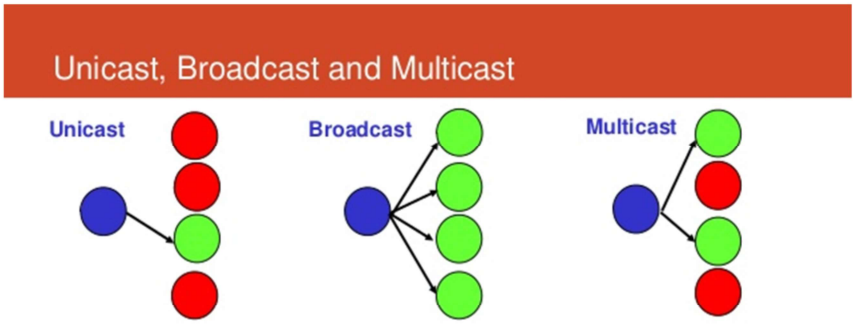
OSI Layer 3: Network Layer



© 2018 Al-Nafi. All Rights Reserved.

The network layer moves data between networks as packets by means of logical addressing schemes.

Unicast, Multicast, and Broadcast Transmissions



- Broadcast: One sender, all the others as receivers
- Unicast: One sender and one receiver
- Multicast: One sender (potentially many senders), many receivers

© 2018 Al-Nafi. All Rights Reserved. 2

In many cases, computer transmission methodology reflects some of the norms that happen in a verbal conversation. Typically, if you want to have a private conversation with an individual, you will take that person aside and speak one-to-one.

A **unicast** is a one-to-one communication between hosts.
If you need to let a group within a crowd of people know about a matter, you can open your announcement with a relevant statement to capture that groups attention within the crowd.

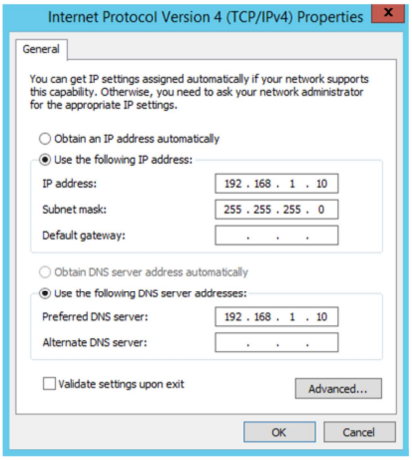
A **multicast** is a one-to-many communication between hosts. If there is something that everyone within a crowd of people should know, such as the need to escape a fire, you wouldn't walk up to each individual and tell them one at a time, you would shout it out for all to hear.

A **broadcast** is a one-to-all communication between hosts. A host can send a broadcast to everyone on its network or sub-network. Depending on the

network topology, the broadcast could have anywhere from one to tens of thousands of recipients. Like a person standing on a soapbox, this is a noisy method of communication. Typically, only one or two destination hosts are interested in the broadcast; the other recipients waste resources to process the transmission. However, there are productive uses for broadcasts. Consider a router that knows a device's IP address but must determine the device's media access control (MAC) address. The router will broadcast an Address Resolution Protocol (ARP) request asking for the device's MAC address.

Multicasting was designed to deliver a stream to only interested hosts. Radio broadcasting is a typical analogy for multicasting. To select a specific radio show, you tune a radio to the broadcasting station. Likewise, to receive a desired multicast, you join the corresponding multicast group. Multicast agents are used to route multicast traffic over networks and administer multicast groups. Each network and sub-network that supports multicasting must have at least one multicast agent. Hosts use Internet Group Management Protocol (IGMP) to tell a local multicast agent that it wants to join a specific multicast group. Multicast agents also route multicasts to local hosts that are members of the multicast's group and relay multicasts to neighboring agents. When a host wants to leave a multicast group, it sends an IGMP message to a local multicast agent. Multicasts do not use reliable sessions; therefore, the multicasts are transmitted as best effort with no guarantee that datagrams are received.

Internet Protocol (IPv4) Networking



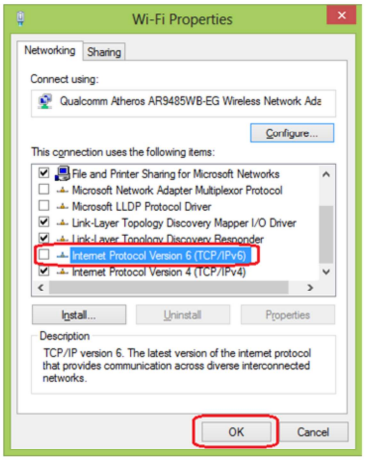
© 2018 Al-Nafi. All Rights Reserved.

The Internet Protocol (IP) is the dominant protocol that operates at the OSI Network Layer 3. IP is responsible for addressing packets so that they can be transmitted from the source to the destination hosts. Because it is an unreliable protocol, it does not guarantee delivery. IP will subdivide the message into fragments when they are too large for a packet. Hosts are distinguished by the IP addresses. The address is expressed as four octets separated by a dot (.), for example, 216.12.146.140. Each octet may have a value between 0 and 255. However, 0 and 255 are not used for hosts. 255 is used for broadcast addresses, and the 0's meaning depends on the context in which it is used. Each address is subdivided into two parts: the network number and the host. The network number assigned by an external organization, such as the Internet Corporation for Assigned Names and Numbers (ICANN), represents the organization's network. The host represents the network interface within the network. The part of the address that represents the network number defines the network's class. Class A network used the leftmost octet as the network number, Class B used the leftmost two octets, etc. The part of the address that is not used as the network number is

used to specify the host. For example, the address 216.12.146.140 represents a Class C network. Therefore, the network portion of the address is represented by the 216.12.146, and the unique host address within the network block is represented by 140. 127, which is the Class A network address block, is reserved for a computer's loopback address. Usually, the address 127.0.0.1 is used. The loopback address is used to provide a mechanism for self diagnosis and troubleshooting at the machine level. This mechanism allows a network administrator to treat a local machine as if it were a remote machine, and ping the network interface to establish whether it is operational.

To ease network administration, networks are typically subdivided into subnets. Because subnets cannot be distinguished with the addressing scheme discussed so far, a separate mechanism, the subnet mask, is used to define the part of the address that is used for the subnet. Bits in the subnet mask are 1 when the corresponding bits in the address are used for the subnet. The remaining bits in the mask are 0. For example, if the leftmost three octets (24 bits) are used to distinguish subnets, the subnet mask is 11111111 11111111 11111111 00000000. A string of 32 1s and 0s is very unwieldy, so the mask is usually converted to decimal notation: 255.255.255.0. Alternatively, the mask is expressed with a slash (/) followed by the number of 1s in the mask. The above mask would be written as /24.

Internet Protocol (IPv6) Networking

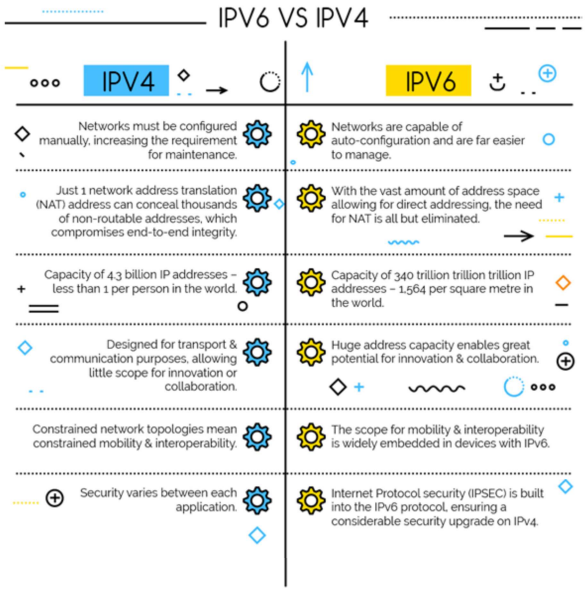


© 2018 Al-Nafi. All Rights Reserved.

IPv6 is a modernization of IPv4 that includes the following:

- A much larger address field: IPv6 addresses are 128 bits, which supports 2128 hosts. Suffice it to say that we will not run out of addresses.
- Improved security: IPSec can be implemented in IPv6. This will help ensure the integrity and confidentiality of IP packets and allow communicating partners to authenticate with each other.
- Improved quality of service (QoS): This will help services obtain an appropriate share of a network's bandwidth.

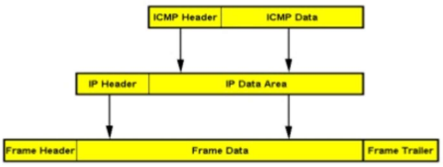
IPv4 vs IPv6



© 2018 Al-Nafi. All Rights Reserved.

Internet Control Message Protocol (ICMP)

Internet Control Message Protocol ICMP



The Internet Control Message Protocol (ICMP) is a control protocol that is considered to be an integral part of IP, although it is architecturally layered upon IP - it uses IP to carry its data end-to-end. ICMP provides error reporting, congestion reporting, and first-hop router redirection.

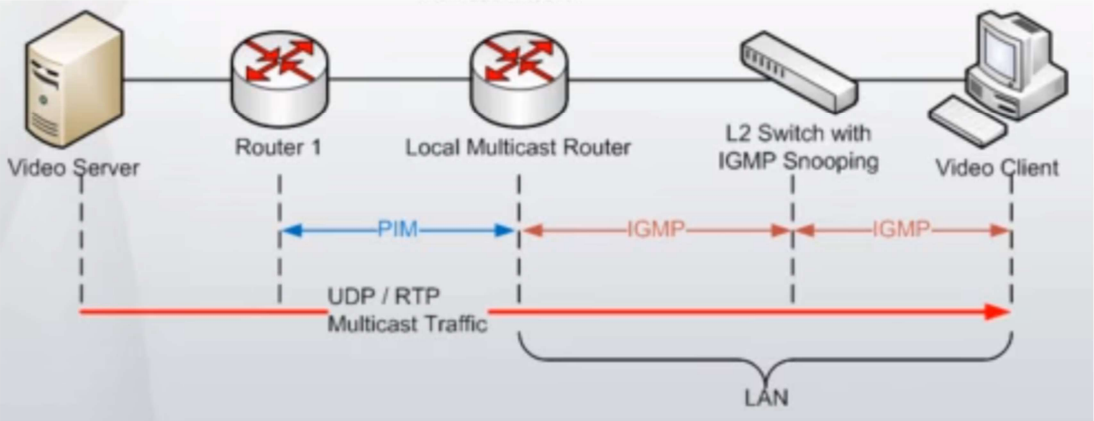
1

© 2018 Al-Nafi. All Rights Reserved.

6

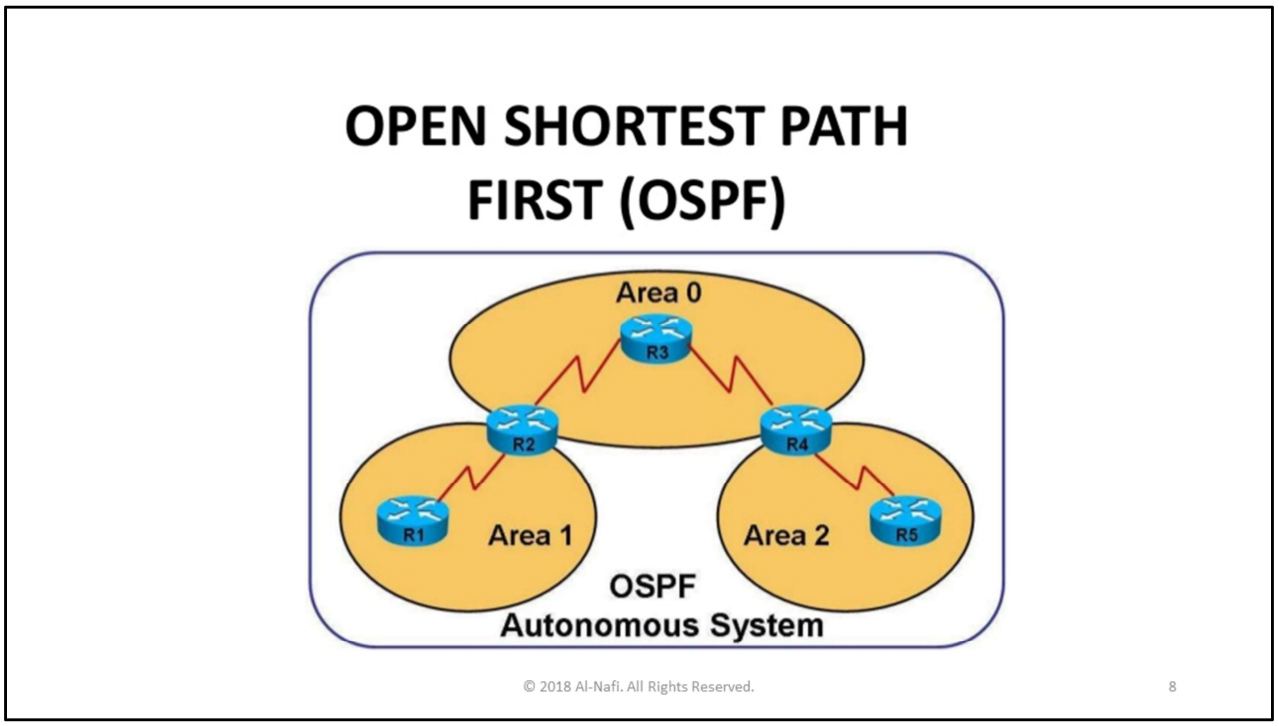
The ICMP is used for the exchange of control messages between hosts and gateways and is used for diagnostic tools such as ping and traceroute. ICMP can be leveraged for malicious behavior, including man-in-the-middle and denial-of-service attacks.

Internet Group Management Protocol (IGMP)



© 2018 Al-Nafi. All Rights Reserved.

IGMP is used to manage multicasting groups that are a set of hosts anywhere on a network that are listening for a transmission. Multicast agents administer multicast groups, and hosts send IGMP messages to local agents to join and leave groups.

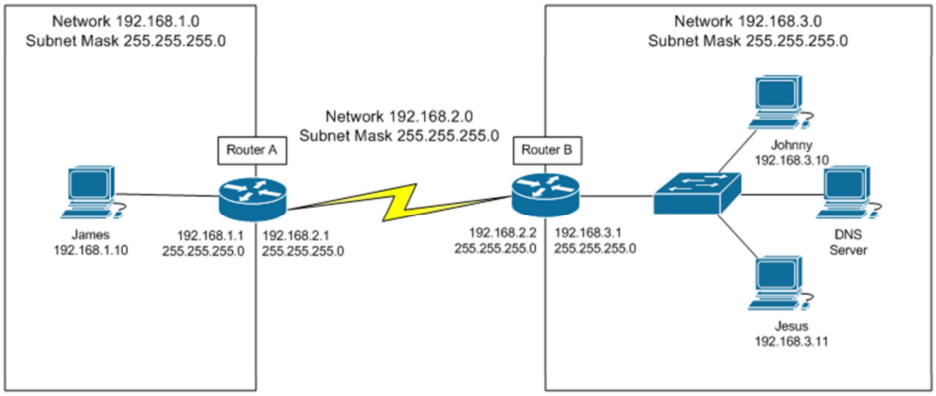


Open Shortest Path First (OSPF) is an interior gateway routing protocol developed for IP networks based on the shortest path first or link-state algorithm. A link-state algorithm can keep track of a total "cost" to calculate the most efficient way of moving information from a source to destination. While a distance vector protocol, such as Routing Information Protocol (RIP), will basically use the number of hops or count of links between networks to determine the best path, a link-state algorithm can surmise the most efficient path by knowing the connecting speed, congestion of the link, availability of the link, and the total hops to determine what might be the best path. A longer hop count could be the shortest path if all other measurements are superior to a path with a shorter hop count.

Routers use link-state algorithms to send routing information to all nodes in an internetwork by calculating the shortest path to each node based on a topography of the internet constructed by each node. Each router sends that portion of the routing table (keeps track of routes to network destinations) that describes the state of its own links, and it also sends the complete

routing structure (topography). The advantage of shortest path first algorithms is that their use results in smaller, more frequent updates everywhere. They converge quickly, thus preventing such problems as routing loops and Count-to-Infinity (when routers continuously increment the hop count to a network). The disadvantage of shortest path first algorithms is that they require substantial amounts of CPU power and memory.

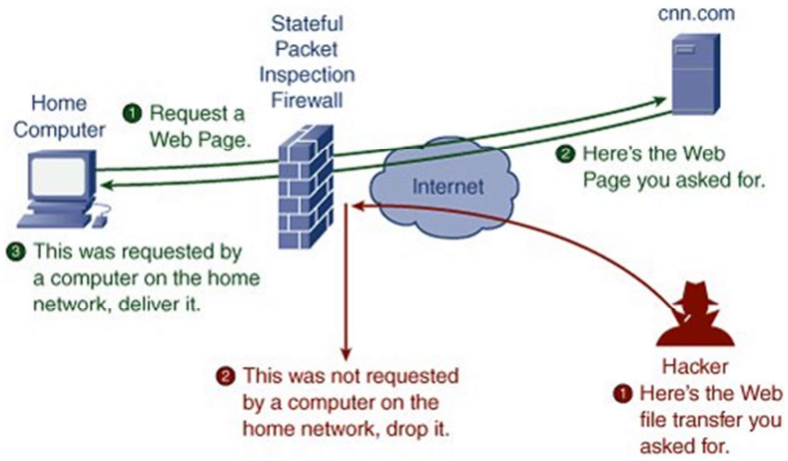
Routers



© 2018 Al-Nafi. All Rights Reserved.

Routers route packets to other networks and are commonly referred to as the Gateway. They read the IP destination in received packets, and based on the router's view of the network, it determines the next device on the network (the next hop) to send the packet. If the destination address is not on a network that is directly connected to the router, it will send the packet to the gateway of last resort, another connected router, and rely on that router to establish a path. Routers can be used to interconnect different technologies and change the architecture. For example, connecting a Token Ring and Ethernet networks to the same router would allow IP Ethernet packets to be forwarded to a Token Ring network. Routers are most commonly used today to connect LANs to
WANs. To build a network, you need switches for the LAN and a router to connect the LAN to the WAN. The most basic security that can be performed at layer 3 on a router is an access control list (ACL) that can define permitted and denied source and destination addresses and ports or services.

Firewalls



© 2018 Al-Nafi. All Rights Reserved.

Routers and firewalls are devices that enforce administrative security policies by filtering incoming traffic based on a set of rules. While a firewall should always be placed at internet gateways, there are also internal network considerations and conditions where a firewall would be employed, such as network zoning. Additionally, firewalls are also threat management appliances with a variety of other security services embedded, such as proxy services and intrusion prevention services (IPS) that seek to monitor and alert proactively at the network perimeter.

Threats and Countermeasures

Technology	Utilization	Threats	Countermeasures
Firewall	Prevent unauthorized access to network resources.	Skilled hackers, misconfigured devices, version/release/update level vulnerabilities.	Schedule and install regular updates and patches. Provide proper training for configuration, maintenance, and operation.
Router	Transmits packets between discreet networks.	Undesired receipt or transmission of data between networks.	Create ACL on router interface to allow or deny IP addresses and services. Augment protection with firewalls, Intrusion Detection and Intrusion Prevention Devices.

© 2018 Al-Nafi. All Rights Reserved.

Threats and Countermeasures continued..

Technology	Utilization	Threats	Countermeasures
ICMP	Verify that a system is responsive running IP.	Smurf: ICMP Echo Request sent to the network broadcast address of a spoofed victim causing all nodes to respond to the victim with an Echo Reply.	Disable ICMP Echo Request on Network.
		Traceroute Exploitation: Traceroute is a diagnostic tool that displays the path a packet traverses between a source and destination host. Traceroute can be used maliciously to map a victim network and learn about its routing.	Disable node enabled deterministic route path; "no ip source-route."
		Ping of Death: Exceeds maximum packet size and causes receiving system to fail.	Redundant and diverse network paths to provide for availability during attack. Disable ICMP Echo Request on Network.
		Ping Scanning: Network mapping technique to detect if host replies to a ping, then the attacker knows that a host exists at that address.	Disable ICMP Echo Request on Network.

© 2018 Al-Nafi. All Rights Reserved.

Threats and Countermeasures continued..

Technology	Utilization	Threats	Countermeasures
IP Fragmentation	Designed to allow units of information to be disassembled or fragmented with the benefit of being delivered in smaller units.	Tear Drop Attack: Exploits the reassembly of fragmented IP packets in the fragment offset field that indicates the starting position, or offset, of the data contained in a fragmented packet relative to the data of the original unfragmented packet. System crashes with accumulation of multiple malformed packets.	Host OS and routers have patching that inspects discrepancy in fragment offset and drops malformed fragment packets.

© 2018 Al-Nafi. All Rights Reserved.